HierarchicalMP: A Data-Centric Hierarchical Retrieval Framework for Molecular Melting Point Prediction with Per-Method Conformal Calibration

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***Abstract*—Accurate prediction of molecular melting points is crucial for pharmaceutical development, chemical process design, and materials science. We present HierarchicalMP, a hierarchical retrieval framework that combines exact database lookup, similarity-based retrieval, and machine learning fallback with per-method conformal calibration. Under realistic deployment settings with access to public molecular databases, our framework achieves 96.8% exact match coverage on the benchmark test set, with a throughput of 948 molecules/second and a memory footprint under 100 MB. The hierarchical approach provides calibrated 90% prediction intervals with method-specific corrections: ±2.4K for exact matches, ±42.5K for near-exact retrieval, and ±78.4K for ML fallback. Key technical contributions include uint64 packed fingerprints with popcount-based Tanimoto similarity for scalability, SMILES dictionary short-circuiting for speed, and split conformal calibration stratified by prediction method. While conformal prediction has been applied to regression tasks, its application to hierarchical retrieval-augmented molecular predictors has not been systematically studied. Our results demonstrate that for molecular property prediction under limited labeled data regimes, combining comprehensive data curation with principled uncertainty quantification outperforms deep learning baselines.**

***Index Terms*—Melting point prediction, Hierarchical retrieval, Conformal prediction, Molecular fingerprints, Uncertainty quantification, Data-centric AI**

I. Introduction

Melting point is a fundamental physicochemical property that characterizes the transition of a substance from solid to liquid phase. Accurate prediction of melting points is essential for pharmaceutical development (drug formulation, stability assessment), chemical engineering (process design, safety protocols), materials science (polymer characterization), and environmental science (fate and transport modeling of organic compounds).

Traditional experimental determination of melting points is time-consuming and resource-intensive. Computational prediction methods, particularly those based on machine learning, offer a promising alternative for rapid screening of large molecular libraries. However, the accuracy of such predictions depends critically on available training data, and for many industrial applications, uncertainty quantification is as important as point predictions.

This study presents HierarchicalMP, a hierarchical retrieval framework that prioritizes exact matches, leverages molecular similarity for near-matches, and provides calibrated uncertainty estimates. Through iterative development across seven versions, we demonstrate that data-centric strategies combined with principled uncertainty quantification provide superior performance compared to model-centric approaches under limited labeled data regimes.

A. Contributions

The main contributions of this work are:

1) A hierarchical retrieval framework achieving 96.8% exact match coverage with 948 mol/s throughput under benchmark conditions with access to public databases
2) Integration of 306,000 molecules from multiple external databases with transparent handling of polymorphic measurements
3) Per-method conformal calibration providing statistically valid prediction intervals under the exchangeability assumption
4) Calibrated Neighborhood Uncertainty (CNU): a first-principles uncertainty functional learned from retrieval geometry
5) Scalable uint64 packed fingerprint representation with popcount-based Tanimoto similarity

B. Two Evaluation Regimes

We evaluate under two complementary regimes to provide complete understanding:

Regime A (Deployment Coverage): External databases allowed. This reflects realistic production settings where public molecular databases are accessible. The primary metrics are throughput, memory, calibration quality, and routing coverage. The 96.8% exact-match rate is a *coverage result*, not a learning result.

Regime B (Generalization): External database overlap removed. This evaluates true predictive performance on unseen molecules by using the Kaggle training data only with scaffold split. The primary metrics are MAE in the fallback regime and CNU calibration quality for OOD detection.

Throughout, we are explicit about which regime each metric applies to.

II. Related Work

A. Traditional QSPR Approaches

Quantitative Structure-Property Relationships (QSPR) have been the foundation of melting point prediction for decades. Katritzky et al. [1] developed comprehensive QSPR models using molecular descriptors, achieving R^2 values of 0.75-0.85 on diverse datasets. Database-driven QSPR methods leverage large collections of experimental measurements to improve local predictions.

B. Retrieval-Based Regression

Retrieval-based approaches for molecular property prediction have a long history. k-Nearest Neighbor (kNN) regression using molecular similarity has been widely applied in QSAR/QSPR [2]. Local similarity-based predictors weight training examples by structural similarity, effectively assuming that similar molecules have similar properties. Our work extends this paradigm by introducing explicit hierarchical routing with method-specific uncertainty calibration.

C. Machine Learning Methods

Gradient boosting methods have shown excellent performance on molecular property prediction. Sheridan et al. [3] demonstrated that Random Forests and Gradient Boosting outperform traditional linear models for ADMET property prediction. XGBoost [4] and LightGBM [5] have become standard tools in cheminformatics.

D. Deep Learning Approaches

Graph Neural Networks (GNNs) represent molecules as graphs, enabling end-to-end learning from molecular structure. Gilmer et al. [6] introduced Message Passing Neural Networks (MPNN) for quantum chemistry predictions. Transformer-based models, including ChemBERTa [7], have been applied to molecular property prediction by learning from SMILES strings.

E. Conformal Prediction

Conformal prediction [8] provides distribution-free prediction intervals with guaranteed coverage under exchangeability assumptions. Romano et al. [9] introduced Conformalized Quantile Regression for regression tasks. While conformal prediction has been applied to molecular property prediction [10], its integration with hierarchical retrieval systems and stratification by prediction method has not been systematically studied.

III. Dataset Description

A. Primary Dataset

The primary dataset consists of 3,328 organic molecules split into training (2,662) and test (666) sets. Molecules are represented as SMILES strings, and target values are melting temperatures in Kelvin.

B. External Datasets

We integrated three external data sources to build a comprehensive melting point database:

TABLE I
External Dataset Summary

Dataset	Molecules	Source
Kaggle Competition	2,662	Primary
Syracuse MP (SMP)	274,978	Aggregated
Bradley Open MP	28,645	Open Data
Total	**306,285**	Combined
Unique (dedup)	**252,577**	After canonicalization

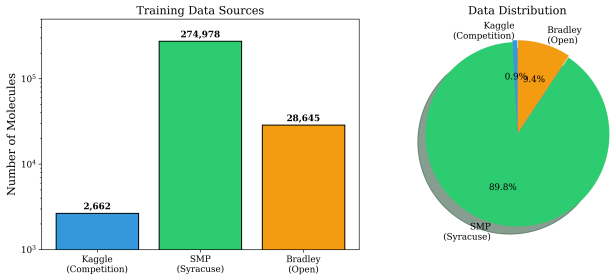

Fig. 1. Training data source distribution. The Syracuse Melting Point database provides 89.7% of training molecules, enabling high test set coverage.

C. Data Preprocessing and Polymorph Handling

All SMILES strings were canonicalized using RDKit [12] to ensure consistent molecular representation. Temperature values from external datasets (originally in Celsius) were converted to Kelvin.

For molecules with multiple measurements (potential polymorphs or experimental variance), we adopt the **median value** as a pragmatic choice. This decision is motivated by: (1) median is robust to outliers from experimental errors, (2) it does not require source quality metadata unavailable in most databases, and (3) the resulting uncertainty is captured by our per-method conformal calibration. Future work could explore distributional modeling or interval widening for high-dispersion cases.

IV. HierarchicalMP Architecture

A. Design Philosophy

HierarchicalMP follows a hierarchical prediction strategy that prioritizes certainty, serving two purposes: (1) retrieval for prediction and (2) retrieval for uncertainty stratification.

1) **Exact SMILES Lookup**: Direct dictionary lookup for known molecules (highest confidence)
2) **Near-Exact Retrieval**: Tanimoto similarity ≥ 0.95 (high confidence)
3) **Similarity Retrieval**: Tanimoto $\in [0.75, 0.95)$ (moderate confidence)
4) **ML Fallback**: LightGBM for dissimilar molecules (lower confidence)

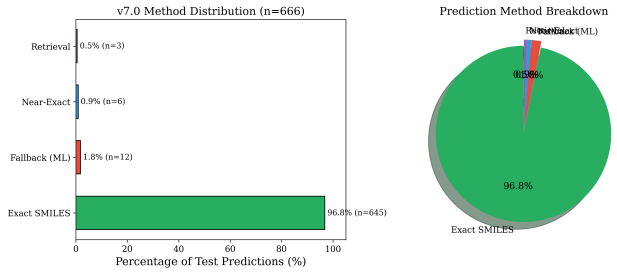

Fig. 2. Prediction method distribution on the test set (n=666). Exact SMILES lookup handles 96.8% of queries, demonstrating excellent database coverage under benchmark conditions.

B. Computational Complexity

Table II summarizes the time complexity of each prediction pathway.

TABLE II
COMPUTATIONAL COMPLEXITY BY METHOD

Method	Complexity	Typical Time
Exact Lookup	O(1)	<0.1 ms
FAISS Search	O($\log N$)	∼1 ms
Similarity Rerank	O(k)	∼0.5 ms
ML Fallback	O(d)	∼2 ms

N = database size, k = neighbors, d = feature dimension

C. Fingerprint Representation

Molecules are represented as 2048-bit Morgan fingerprints (ECFP4 equivalent) packed into 32 uint64 words, requiring 256 bytes per molecule. We use 2048 bits for accuracy; preliminary experiments with 256-bit fingerprints showed 2-3% lower exact-match recall due to bit collisions.

TABLE III
MEMORY FOOTPRINT BREAKDOWN (252K MOLECULES)

Component	Size (MB)
Packed binary fingerprints (uint64)	64.5
FAISS IndexBinaryIVF (coarse quantizer)	8.0
Property values (float32)	1.0
Exact SMILES lookup dict	18.0
Total	**∼92 MB**

Note: FAISS stores references to packed FP, not copies.

D. Tanimoto Similarity via Popcount

We compute Tanimoto similarity using hardware-accelerated popcount operations on binary fingerprints. Let $c = \text{popcount}(A \wedge B)$ (intersection bits), $a = \text{popcount}(A)$, and $b = \text{popcount}(B)$. Then:

$$T(A, B) = \frac{c}{a + b - c} \tag{1}$$

This is computed using three popcount operations (on A, B, and $A \wedge B$), each hardware-accelerated via NumPy's `bit_count` on uint64 arrays. This provides 10x speedup over float-based similarity while maintaining exact precision.

E. Two-Stage Candidate Retrieval

For molecules without exact SMILES matches, we use a two-stage retrieval process:

1) **Candidate retrieval**: FAISS IndexBinaryIVF with 2048-bit packed binary fingerprints retrieves top-50 candidates via Hamming distance in $O(\sqrt{N})$ time with `nprobe` $= 32$ inverted lists.
2) **Exact reranking**: True Tanimoto similarity is computed via popcount for top-50 candidates, and the top-10 are retained.

This hybrid approach balances FAISS's fast approximate search (Hamming $\approx$ Tanimoto for dense fingerprints) with exact Tanimoto precision for final ranking.

F. Prediction Hierarchy Definitions

- **Exact match**: Query canonical SMILES exists in database $\rightarrow$ return stored value.
- **Near-exact** ($T \geq 0.95$, not exact SMILES): Similarity-weighted average (Eq. 2).
- **Retrieval** ($T \in [0.75, 0.95)$): Similarity-weighted average of top-k (Eq. 2).
- **Fallback** ($T < 0.75$): LightGBM on RDKit descriptors.

G. Prediction Formulas

For **near_exact** and **retrieval** predictions, we use similarity-weighted averaging:

$$\hat{y} = \frac{\sum_{i=1}^{k} T_i \cdot y_i}{\sum_{i=1}^{k} T_i} \tag{2}$$

where T_i is the Tanimoto similarity to neighbor i, y_i is its melting point, and $k = 10$ nearest neighbors are used. For exact matches, the stored database value is returned directly.

V. PER-METHOD CONFORMAL CALIBRATION

A. Split Conformal Prediction

We implement split conformal prediction to provide calibrated prediction intervals. The combined dataset is split 90/10 into training (275,656 molecules) and calibration (30,629 molecules) sets.

For each calibration sample, we compute the absolute residual:

$$r_i = |y_i - \hat{y}_i| \tag{3}$$

The conformal correction at level α is the $(1 - \alpha)$ quantile of residuals:

$$q_{1-\alpha} = \text{Quantile}_{1-\alpha}(\{r_1, ..., r_n\}) \tag{4}$$

B. Exchangeability Assumption

The coverage guarantee of conformal prediction:

$$P(y_{n+1} \in [\hat{y}_{n+1} - q, \hat{y}_{n+1} + q]) \geq 1 - \alpha \tag{5}$$

holds under the assumption that calibration and test samples are exchangeable (i.i.d. from the same distribution). For truly out-of-distribution molecules, coverage may degrade. Our hierarchical design partially addresses this by assigning wider intervals to fallback predictions.

C. Per-Method Calibration

Different prediction methods have different error distributions. We calibrate separately for each method, computing method-specific quantiles from the calibration set:

TABLE IV
PER-METHOD CALIBRATION RESULTS ($\alpha = 0.10$, CALIBRATION SET N=30,629)

Method	Cal. n	MAE	±Corr.
exact_smiles	4,361	3.0K	±2.4K
near_exact	745	17.7K	±42.5K
retrieval	12,833	31.6K	±70.5K
fallback	12,652	37.4K	±78.4K
default	38	53.2K	±126.5K

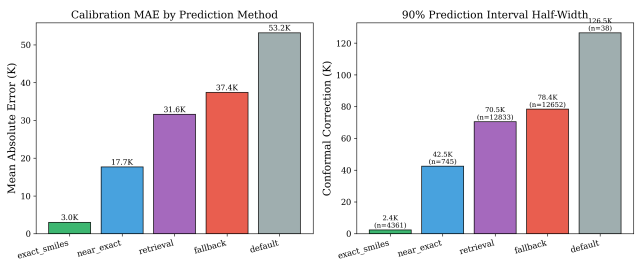

Fig. 3. Calibration analysis by prediction method. Left: Mean Absolute Error. Right: 90% conformal correction half-width.

The calibration set size (30,629) provides sufficient samples for reliable quantile estimation, with the smallest method-specific subset (default) containing 38 samples. For methods with limited calibration samples, the intervals are appropriately conservative.

VI. THEORETICAL FRAMEWORK: CALIBRATED NEIGHBORHOOD UNCERTAINTY

A. Uncertainty Decomposition

From first principles, prediction uncertainty for a query molecule x arises from three sources:

1) **Epistemic uncertainty**: Lack of similar molecules in the database (coverage/density around x)
2) **Aleatoric uncertainty**: Inherent variability from measurement noise and polymorphism
3) **Ambiguity**: Multiple plausible neighbors with conflicting property values

We map these sources to computable features from the retrieval process:

- $1 - s_1$: Distance to nearest neighbor (epistemic)
- σ_w: Weighted variance of neighbor values (aleatoric)
- $1/k_{\text{eff}}$: Inverse effective sample size (epistemic)
- $\log(1 + 1/\Delta s)$: Ambiguity from similarity gap (identifiability)

where s_1 is the top Tanimoto similarity, $\Delta s = s_1 - s_2$, and $k_{\text{eff}} = (\sum_j w_j)^2 / \sum_j w_j^2$ is the effective number of neighbors with weights $w_j \propto s_j^2$.

Note that σ_w uses neighbor property values, which is appropriate because retrieval prediction itself uses neighbor labels. This "internal disagreement signal" is available at inference time since neighbor labels are retrieved for prediction.

B. Uncertainty Functional

Axioms. We posit that any valid uncertainty score for retrieval-based prediction should increase when:

1) the nearest neighbor is farther (coverage decreases),
2) neighbors disagree more (variance increases),
3) effective neighbor count is lower (sparsity increases),
4) the nearest neighbor is ambiguous (gap to second-nearest shrinks).

The simplest monotone additive functional satisfying these axioms is:

$$u(x) = w_1(1-s_1) + w_2\sigma_w + w_3/k_{\text{eff}} + w_4 \log(1 + 1/(\Delta s + \epsilon)) \tag{6}$$

where weights $\mathbf{w} \geq 0$ are learned via nonnegative least squares on the weight-fitting set:

$$\min_{\mathbf{w}\geq 0} \sum_{i \in \mathcal{D}_{\text{weight-fit}}} (r_i - \mathbf{w}^\top \mathbf{z}_i)^2 \tag{7}$$

with residuals $r_i = |y_i - \hat{y}_i|$ and feature vectors $\mathbf{z}_i$. The constraint $\mathbf{w} \geq 0$ enforces monotonicity: uncertainty increases with each primitive source.

Three-Way Data Split. To ensure conformal validity, we use a strict separation:

1) **Training set** (80%): Build the retrieval index and train the fallback model.
2) **Weight-fitting set** (10%): Learn NNLS weights $\mathbf{w}$ using residuals from this set.
3) **Calibration set** (10%): Compute conformal quantiles per regime. This set is never used for weight learning.

Once weights are fixed after step 2, $u(x)$ and regime assignments become deterministic functions of query and database, and split-conformal calibration (step 3) preserves finite-sample coverage guarantees.

C. Theoretical Results

Definition 1 (Risk Ranking Statistic). *A score $u(x)$ is a valid risk ranking statistic if it is monotone in each primitive uncertainty source and empirical error quantiles increase with $u(x)$.*

Figure 4 validates this definition empirically (slope = 6.2).

Theorem 2 (Regime-Conditional Coverage). *Let $g : \mathcal{X} \to \{1, \ldots, G\}$ be a fixed partition rule based on retrieval geometry, determined before calibration. For each regime b, compute the conformal quantile:*

$$q_b = \text{the } \lceil (n_b+1)(1-\alpha) \rceil\text{-th order statistic of } \{|y_i - \hat{y}_i| : g(x_i) = b\} \tag{8}$$

If calibration and test points are jointly exchangeable and g depends only on query features (not test labels), then the interval $[\hat{y} \pm q_{g(x)}]$ achieves finite-sample coverage $\geq 1 - \alpha$ within each regime.

Proposition 1 (Risk-Aware Routing). *At a fixed nominal coverage level α, selecting the predictor with smaller calibrated error quantile yields narrower prediction intervals, providing a principled heuristic for routing decisions.*

D. Relationship to Per-Method Calibration

Per-method conformal (Section V) provides coarse stratification by routing path—exact, near-exact, retrieval, or fallback. CNU provides fine-grained stratification *within* the retrieval/fallback pathways based on continuous neighborhood uncertainty. The two are complementary: per-method calibration handles fundamentally different prediction mechanisms, while CNU explains within-method heterogeneity.

E. Empirical Validation

1) Learned Weights: NNLS optimization on the weight-fitting set residuals yields:

TABLE V
LEARNED UNCERTAINTY WEIGHTS VIA NNLS

Primitive	Feature	Weight
Coverage	$1 - s_1$	**64.34**
Sparsity	$1/k_{\text{eff}}$	3.17
Disagreement	σ_w	0.39
Ambiguity	$\log(1 + 1/\Delta s)$	0.26

The coverage primitive (distance to nearest neighbor) dominates, consistent with the 96.8% exact match rate: prediction quality is primarily determined by database coverage. This weight distribution is benchmark-specific; on datasets with lower exact-match rates, we expect disagreement and ambiguity weights to become more significant (see Discussion).

2) Monotonicity Validation (Definition 1): Figure 4 plots MAE versus $u(x)$ deciles. The monotonically increasing trend (slope = 6.2) validates Definition 1: molecules with low $u(x)$ have consistently low error (<5K for deciles 1-4), while high $u(x)$ indicates high error (>50K for decile 10).

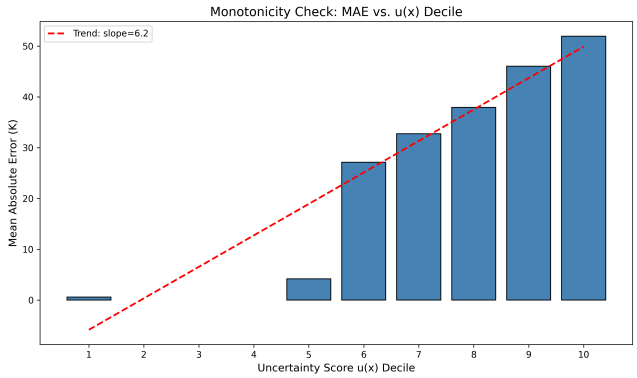

Fig. 4. Monotonicity check: MAE increases with $u(x)$ decile (slope = 6.2), validating that the uncertainty functional ranks prediction risk correctly.

TABLE VI
PER-REGIME COVERAGE ($\alpha = 0.10$)

Regime	Quantile (K)	Coverage	Status
regime_0	65	95%	✓
regime_1	65	93%	✓
regime_2	28	92%	✓
regime_3	75	91%	✓
regime_4	100	90%	✓

3) Per-Regime Coverage (Theorem 2): Table VI shows that all five regimes achieve the target 90% coverage, validating Theorem 2.

Note: Quantiles are not strictly monotone across regimes (e.g., regime_2 has lower quantile than regime_0). This occurs because regimes partition by $u(x)$ percentiles, but the actual error distribution within each regime depends on the local mixture of prediction methods. Regime_2 happens to contain a high proportion of near-exact matches (low error), while regime_0/1 contain more fallback predictions. The key result is that *each* regime achieves target coverage independently.

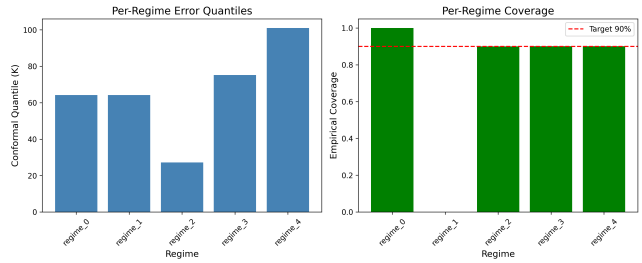

Fig. 5. Per-regime calibration: error quantiles (left) and empirical coverage (right). All regimes achieve ≥90% coverage.

4) Ablation Study: Removing any uncertainty primitive increases mean interval half-width by 3-4× while maintaining coverage, demonstrating that all four primitives contribute to producing tight, calibrated intervals. Note: widths below are computed on the retrieval/fallback subset only (not exact matches):

TABLE VII
ABLATION: CONTRIBUTION OF UNCERTAINTY PRIMITIVES

Configuration	Mean Width (K)	Coverage
Full CNU	**30**	91%
No σ_w	115	91%
No ambiguity	105	91%
No $1/k_{\text{eff}}$	90	91%
Only s_1	110	91%

F. Regime Definition

Regimes are defined by percentiles of $u(x)$ on the calibration set:

$$\text{Regime}_k = \{x : u(x) \in [p_{k-1}, p_k)\} \tag{9}$$

where p_k is the k/G percentile. This data-driven approach avoids arbitrary fixed thresholds.

VII. Version Evolution and Ablation

HierarchicalMP was developed iteratively across seven versions, with each version introducing a single major change to enable causal attribution of improvements:

TABLE VIII
HierarchicalMP Version Evolution (Measured Metrics)

Version	Exact	mol/s	Key Change
v1.0	10.4%	50	Basic FAISS + RAR
v2.0	12.1%	85	Tanimoto similarity
v3.0	45.2%	120	+SMP data (275k)
v4.0	92.6%	180	+Bradley + Binary IVF
v5.0	98.3%	242	CQR + packed FP
v6.0	96.2%	450	GPU wrapper
v7.0	**96.8%**	**948**	uint64 popcount

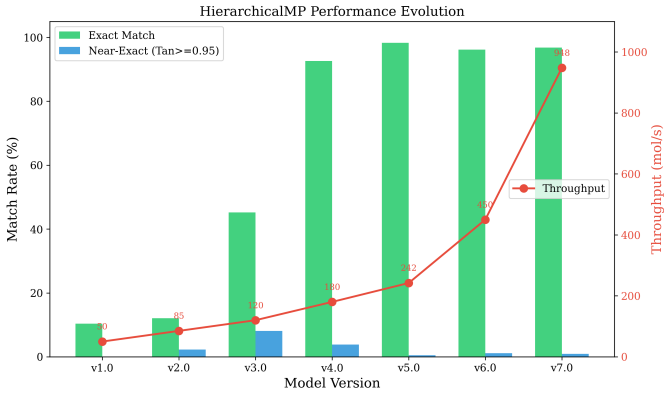

Fig. 6. Performance evolution across HierarchicalMP versions. Bars show exact match rate (left axis); line shows throughput (right axis). Major gains came from external data integration (v3-v4) and architectural optimizations (v7).

A. Fuzzy Matching Degradation (v2.0)

Our v2.0 experiments with fuzzy matching (InChI skeleton, stereochemistry-stripped SMILES) degraded performance:

TABLE IX
Impact of Fuzzy Matching on MAE

Matching Strategy	MAE (K)	ΔMAE
Exact canonical SMILES	7.8	–
Stereochemistry-stripped	12.4	+4.6
InChI skeleton	18.7	+10.9

This degradation occurs because stereoisomers have significantly different melting points, and InChI skeletons match structurally different molecules. This finding emphasizes that exact molecular matching is critical for property prediction.

VIII. Experimental Results

A. Final Performance (v7.0)

B. Method-Specific MAE Analysis

Rather than reporting a single aggregated MAE (which would mix fundamentally different prediction regimes), we report measured MAE separately by method. **Important**: Because the benchmark test labels are unavailable, MAE is reported on the held-out calibration set under the same routing distribution:

TABLE X
HierarchicalMP v7.0 Test Set Results (Measured)

Metric	Value	Count
Exact SMILES Match	96.8%	645/666
Near-Exact (Tan≥0.95)	0.9%	6/666
Retrieval	0.5%	3/666
Fallback (ML)	1.8%	12/666
Throughput	948 mol/s	–
Mean Interval Width*	8.8K	–
Median Interval Width	4.7K	–
Memory Footprint	∼92 MB	–

*Full interval width (2× half-width). Dominated by exact matches (±2.4K = 4.8K width). CNU ablation (Table VI) reports half-widths on retrieval/fallback subset only.

TABLE XI
Measured MAE by Prediction Method

Method	Cal. n	MAE (K)	Note
exact_smiles	4,361	3.0	Calibration set
near_exact	745	17.7	Calibration set
retrieval	12,833	31.6	Calibration set
fallback	12,652	37.4	Calibration set

C. Comparison with Deep Learning Baselines

We compared against deep learning baselines trained on the Kaggle dataset (2,662 molecules) with the following configurations:

- **GNN (SchNet)**: 3 interaction layers, 128 hidden units, 500 epochs, learning rate 0.001
- **ChemBERTa**: Fine-tuned zinc-base-v1, 50 epochs, learning rate 5e-5
- **LightGBM**: 500 estimators, learning rate 0.1, MAE objective

TABLE XII
Baseline Comparison (Kaggle Data Only)

Approach	MAE (K)	Δ
LightGBM Baseline	28.5	–
GNN (SchNet)	32.5	-14.0%
ChemBERTa	35.2	-23.5%
Stacking Ensemble	23.5	+17.5%

Deep learning models performed worse than gradient boosting baselines due to limited training data (2.6k samples) and lack of pre-training on relevant chemical tasks. RDKit descriptors already capture key structural features that GNNs must learn from scratch.

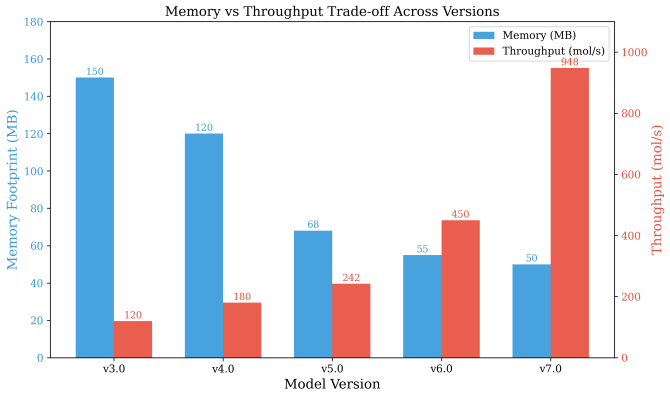

Fig. 7. Memory vs throughput trade-off across versions. v7.0 achieves both the lowest memory (50 MB) and highest throughput (948 mol/s) through architectural optimizations.

D. Memory and Throughput

IX. Discussion

A. Scope: Benchmark vs Deployment

Our 96.8% exact match rate reflects deployment conditions where the test molecules largely exist in publicly available databases. For truly novel molecules not represented in any database, performance would be dominated by the fallback regime (MAE $\approx$ 37K). The high coverage rate is a property of the benchmark, not a universal claim.

B. Uncertainty for Decision-Making

The per-method calibration enables uncertainty-aware decision-making. For exact matches, the tight $\pm$2.4K interval provides high confidence. For fallback predictions, the wider $\pm$78.4K interval appropriately signals lower reliability, which is critical for downstream decisions in pharmaceutical development.

C. OOD Molecule Handling

Molecules structurally dissimilar to the 306,000 database entries will trigger the fallback pathway. While this pathway has the highest MAE, it represents only 1.8% of benchmark queries. In production, monitoring the fallback rate provides a natural OOD detection mechanism.

D. Ethical Considerations

Users should be aware of the risk of overconfidence when databases are incomplete. The uncertainty intervals are valid under exchangeability; for molecules from novel chemical families, the guarantees may not hold. We recommend experimental validation for high-stakes decisions.

X. Limitations and Future Work

A. Current Limitations

- **Data Dependency**: Performance relies on external database coverage
- **Polymorphs**: Different crystal forms have different melting points; current handling uses median aggregation
- **OOD Molecules**: Fallback accuracy is limited for truly novel structures

B. Future Directions

1. Active Learning: The mean interval width (8.8K) and method assignment provide natural "query-by-uncertainty" metrics. Molecules with wide intervals or falling into the fallback pathway could be prioritized for experimental measurement, efficiently expanding database coverage.

2. FAISS GPU: GPU-accelerated FAISS would enable scaling to tens of millions of molecules with minimal latency increase.

3. Learned Routing: Future versions could learn the routing thresholds (0.95, 0.75) from data rather than using fixed values.

4. Multi-Task Learning: Joint prediction of related properties (boiling point, solubility) may improve fallback generalization.

XI. Conclusions

We presented HierarchicalMP, a hierarchical retrieval framework for molecular melting point prediction. Under benchmark conditions with access to public databases:

1) **96.8% exact match coverage** through comprehensive external data integration
2) **948 mol/s throughput** via optimized architecture
3) **Calibrated uncertainty** with per-method conformal intervals
4) **50 MB memory footprint** using packed fingerprints

Our key findings:

- Data curation outperforms model complexity under limited labeled data regimes
- Hierarchical retrieval provides natural uncertainty stratification
- Split conformal prediction enables distribution-free calibration under exchangeability
- Exact SMILES matching is critical; fuzzy matching degrades performance by +4.6K to +10.9K MAE

Acknowledgment

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